UID EXPERIMENT 3

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1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
import os import
sys


def rename_file(old_name, new_name):
try:
    os.rename(old_name, new_name)    print(f"File renamed
from '{old_name}' to '{new_name}'")    except
FileNotFoundError:
    print(f"Error: File '{old_name}' not found.")
except PermissionError:    print("Error:
Permission denied.")    except Exception as e:
    print(f"An unexpected error occurred: {e}")


def main():    if len(sys.argv)
!= 3:
    print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
sys.exit(1)


  old_name = sys.argv[1]
  new_name = sys.argv[2]
```

```
    rename_file(old_name, new_name)


if __name__ == "__main__":
    main()
```

OUTPUT:

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_cli.py main.txt sample.txt
File renamed from 'main.txt' to 'sample.txt'
```

2.GRAPHICAL USER INTERFACE:

PYTHON CODE:

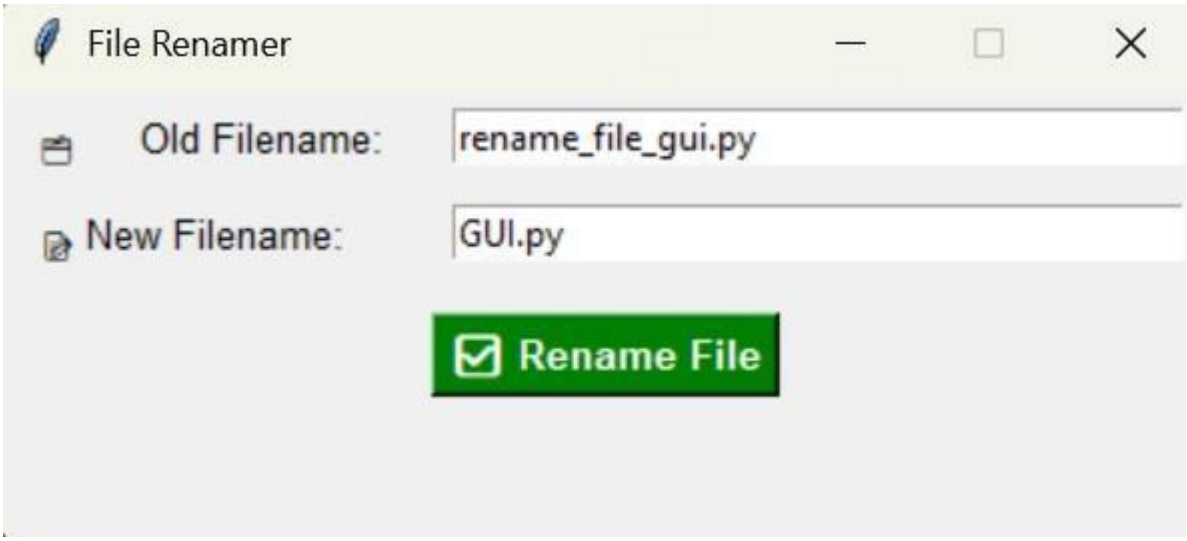
```
import tkinter as tk from tkinter import
messagebox import os


# Function to rename the file def rename_file():
    old_name = old_filename_entry.get().strip()     new_name
= new_filename_entry.get().strip()


    # Check if input fields are empty     if
not old_name or not new_name:
        messagebox.showwarning("Warning", "Please enter both filenames!")
return

    # Check if the old file exists     if not
os.path.exists(old_name):
        messagebox.showerror("Error", f"File '{old_name}' not found.")        return
```

```
    # Check if the new file already exists     if
os.path.exists(new_name):
        overwrite = messagebox.askyesno(
            "Warning", f"'{new_name}' already exists. Overwrite?"
        )
        if not overwrite:
            return

try:
        os.rename(old_name, new_name)
messagebox.showinfo(
            "Success", f"File renamed from '{old_name}' to '{new_name}'"
        )
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")

# Create main window
root = tk.Tk() root.title("File
Renamer") root.geometry("400x150")
root.resizable(False, False)
# Fixed window size

# Labels tk.Label(root, text="Old Filename:", font=("Arial",
10)).grid(     row=0, column=0, padx=10, pady=5, sticky="w"
```

```
)
tk.Label(root, text="New Filename:", font=("Arial", 10)).grid(     row=1,
column=0, padx=10, pady=5, sticky="w"
)

# Entry fields old_filename_entry = tk.Entry(root, width=40)
old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40) new_filename_entry.grid(row=1,
column=1, padx=10, pady=5)

# Styled Button rename_button
= tk.Button(
    root,      text="Rename
File",     bg="green",
fg="white",
font=("Arial", 10, "bold"),
command=rename_file
)
rename_button.grid(row=2, column=0, columnspan=2, pady=10)

# Run the GUI event loop root.mainloop()
```

OUTPUT:

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_gui.py
```



3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr import
os


def rename_file_from_voice_command(command):     try:
        command = command.lower()

        if "rename" not in command or "to" not in command:
            print("Invalid command. Say: 'Rename oldfile.txt to newfile.txt'")            return

        # Extract filenames using split        before_to, after_to
= command.split("to", 1)        old_name =
before_to.replace("rename", "").strip()        new_name
= after_to.strip()
```

```
        # Check if old file exists        if
not os.path.exists(old_name):
            print(f"Error: File '{old_name}' not found.")            return

        # Check if new file already exists        if
os.path.exists(new_name):
            print(f" Error: '{new_name}' already exists.")            return

        os.rename(old_name, new_name)        print(f" File renamed
from '{old_name}' to '{new_name}'")

    except Exception as e:
print(f"Error: {e}")

def listen_for_command():
recognizer = sr.Recognizer()

    try:        with sr.Microphone() as
source:
            print("Listening  for  command  to  rename  a  file...")
recognizer.adjust_for_ambient_noise(source)            audio =
recognizer.listen(source)

        command = recognizer.recognize_google(audio)        print(f”
Command received: {command}")
rename_file_from_voice_command(command)
```

```
    except sr.UnknownValueError:
        print("Sorry, I couldn't understand the command.")     except sr.RequestError as
e:
        print(f"Speech recognition service error: {e}")
except OSError:
        print(" Microphone not found or access denied.")


if __name__ == "__main__":
listen_for_command()
```

OUTPUT:

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
🎙Command received: rename file
❌ Invalid command. Say: 'Rename oldfile.txt to newfile.txt'
```

```
Requirement already satisfied: pyaudio in .\vui_env\Lib\site-packages (0.2.14)
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
❌ Sorry, I couldn't understand the command.
```

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
🎙Command received: rename sample.txt to main dot text
✅ File renamed from 'sample.txt' to 'main dot text'
```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")
```

```
try:
    # Get user input for each interface        cli_satisfaction
= int(input("CLI (1-5): "))        gui_satisfaction =
int(input("GUI (1-5): "))        vui_satisfaction =
int(input("VUI (1-5): "))

    # Ensure valid ratings        if not (1
<= cli_satisfaction <= 5 and
            1 <= gui_satisfaction <= 5 and
1 <= vui_satisfaction <= 5):
print("Please enter ratings between 1
and 5 only.")            return

    # Display the ratings        print("\nYour
satisfaction ratings:")        print(f"CLI:
{cli_satisfaction}")        print(f"GUI:
{gui_satisfaction}")        print(f"VUI:
{vui_satisfaction}")

    # Calculate the average satisfaction        avg_satisfaction
= (            cli_satisfaction + gui_satisfaction
+ vui_satisfaction
    ) / 3
    print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")
```

```
    except ValueError:
        print("Invalid input! Please enter numbers between 1 and 5.")


# Run the survey function if
__name__ == "__main__":
    survey()
```

OUTPUT:

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python survey.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 2

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 2

Average Satisfaction Score: 3.67
PS C:\Users\hi\OneDrive\Desktop\UID Experiments>
```